

Batch Obrada Podataka Pitanja

1. Which 5 stocks have seen the highest percentage rise in a given year?
2. What is the average annual return for each exchange (NASDAQ, NYSE, S&P500)?
3. Which stocks have consistently outperformed their exchange index over the past 5 years?
4. Which stocks exhibit the highest volatility (standard deviation of daily returns)?
5. What is the correlation between trading volume spikes and subsequent price movements?
6. Which **x** stocks have the largest gap between daily high and low in **y month** (intraday volatility)?
7. Which stocks have shown consistent upward trends (positive returns) for 6+ consecutive months?
8. How has the concentration of top 10 stocks changed over time (market dominance)?
9. What is the recovery time for stocks after significant drops (>10% single day)?
10. Which stocks have seen the most significant increase in trading volume over **x** year?

Streaming Obrada Podataka Pitanja

1. Which stocks have the highest price volatility (price swing) in the last 10 minutes?
2. Which stocks are experiencing unusual trading activity (volume spike) compared to their historical daily average? (Stream-Batch Join)
3. What is the trade frequency (number of trades per minute) for each stock, and which are the most actively traded right now?
4. What is the real-time market sentiment (ratio of advancing vs declining stocks) per exchange?
 - a. Window: Tumbling (1 min)
 - b. Logic: Compare current price vs previous window close
 - c. Aggregation: COUNT(advancers) / COUNT(decliners) per exchange
 - d. Output: Market breadth indicator updated every minute
5. What is the average percentage change across all tracked stocks in the last 15 minutes, grouped by exchange?
 - a. Window: Tumbling (15 min)
 - b. Aggregation: AVG(price_change_pct) GROUP BY exchange
 - c. Output: Exchange-level momentum indicator